

Flow is All You Need for WiFi: Rectifying 3D Human Poses without Complex Architectures

1st Nguyen Vinh Nghi, 2nd Le Ngoc Anh Thu, 3rd Nguyen Ngo Nhat Nam, and Tran Ngoc Hoang
FPT University, Ho Chi Minh Campus, Vietnam
{CE182108, CE180615, CE181939}@fpt.edu.vn, and mentor@fe.edu.vn

Abstract

WiFi-based human pose estimation offers an occlusion-resilient alternative to camera-based sensing, but estimating multi-person 3D skeletons from channel state information remains difficult because WiFi signals are noisy, low-dimensional, and strongly affected by multipath propagation. This paper presents a compact draft-to-refine redesign of the Person-in-WiFi 3D pipeline. A controlled architecture ladder separates representation, sequence encoder, pose head, and flow-refinement choices. Under the inherited three-receiver Person-in-WiFi 3D evaluation protocol, Mamba-2 Flow (two steps) achieves 165.487 mm MPJPE: a 7.053 mm (4.09%) improvement over the PETR reference, while increasing throughput from 118.7 to 209.9 FPS, reducing parameters from 13.130M to 3.618M, and reducing peak allocated memory from 155.60 to 25.29 MB. The result establishes compact sequence modeling with two-step flow refinement as an accuracy-oriented alternative under this WiFi pose-estimation protocol.

Index Terms

WiFi sensing, CSI, 3D human pose estimation, Mamba2, flow matching, spectral tokenization.

I. INTRODUCTION

WiFi channel state information (CSI) provides a non-visual sensing signal that can be used for human perception in indoor environments. Unlike camera-based pose estimation, WiFi sensing can operate under occlusion and low-light conditions, but the input is indirect: body motion perturbs multipath propagation rather than producing explicit visual evidence. This makes multi-person 3D pose estimation from WiFi substantially harder than pose estimation from images.

Person-in-WiFi 3D demonstrated an end-to-end PETR-style formulation for multi-person 3D pose estimation from WiFi CSI [1]. We ask whether its pose-decoding path can be made smaller and faster while preserving pose accuracy. The resulting architecture ladder compares spectral tokenization, a PETR query decoder and a lightweight query-pose head, Transformer and Mamba-family encoders, draft-only variants, and one- versus two-step flow refinement. The selected Mamba-2 Flow (two steps) configuration combines spectral tokenization, a flattened Mamba-2 sequence encoder, a lightweight query-pose head, and two-step flow refinement.

The study uses the three-receiver, fixed-layout Person-in-WiFi 3D setup and its temporal train/test split. It therefore evaluates compact architecture choices under a common sensing and evaluation protocol; cross-person and cross-environment generalization are outside the scope of this ablation.

“Flow is All You Need” describes the final correction stage of this compact architecture, rather than flow in isolation. In the studied setting, lightweight flow refinement complements a compact query-pose head while retaining strong pose accuracy.

Our contribution is a controlled architecture study that jointly reports MPJPE, scene-cardinality results, throughput, parameter count, and peak allocated memory. The ablation identifies Mamba-2 Flow (two steps) as the lowest-MPJPE configuration and Mamba-2 Draft as the throughput and memory endpoint; complementary diagnostics characterize both operating points.

II. RELATED WORK

A. Human Pose from Images

Camera-based pose estimation is the most mature paradigm and supplies the conceptual templates this paper inherits [2]. Bottom-up methods such as OpenPose [3] regress per-pixel keypoint heatmaps and part affinity fields and then associate them into instances, which scales gracefully with crowd size. Top-down methods such as AlphaPose [4] first run a person detector and then a single-person estimator, usually attaining higher accuracy at the cost of cropping overhead. High-resolution backbones such as HRNet [5] sharpen the accuracy ceiling by preserving spatial resolution across the network. These methods motivate the set-prediction perspective we later adopt for WiFi but inherit the deployment limits of cameras—appearance capture, lighting dependence, and reduced reliability under occlusion—which conflict with privacy-sensitive indoor monitoring.

B. Human Pose from Wi-Fi

Radio and WiFi sensing developed from through-wall pose recovery and ambient physiological or gesture sensing [6]–[9]. CSI-based representations then supported person perception and pose recovery without an image stream [10]–[13]. WiNect and GoPose advanced free-form single-person 3D tracking [14], [15], while MM-Fi broadened multimodal wireless benchmarks [16]. Application-oriented systems have also considered metaverse avatars, joint activity–pose analysis, and industrial skeleton

monitoring [17]–[19]. Recent graph-based and lightweight continuous-pose studies further illustrate the range of structural and efficiency choices in WiFi pose modeling [20], [21].

Person-in-WiFi 3D adapts the detection-and-regression perspective to WiFi CSI and provides the baseline used in this study [1]. A related short-form study also reports 3D WiFi pose estimation under commodity sensing [22]. Transformer architectures became important for structured prediction after the success of attention-based sequence modeling [23], while the DETR family of set predictors [24] and its pose-estimation extension PETR [25] provide the architectural template we modify.

State space models such as Mamba provide linear-time sequence modeling with selective recurrence [26]. Vision Mamba, VMamba, and LocalMamba show how selective state-space computation can be adapted to structured visual tokens [27]–[29]. Mamba-2 further connects structured state space models and attention-like computation [30]. These properties motivate replacing a heavier Transformer encoder with a compact flattened Mamba-2 sequence encoder for WiFi token sequences.

Rectified flow learns a velocity field for transporting samples between source and target distributions [31]. Diffusion-based 3D pose methods have likewise used iterative generation, multi-hypothesis aggregation, and learned refinement to improve pose structure [32]–[34]. Here, however, the source is not random noise; it is a draft 3D pose produced by the detection head. The flow module is therefore used as a learned pose refinement operator. Because no reflow stage is performed, the interpretation is closer to conditional flow-matching refinement than a full generative rectified-flow pipeline.

Recent WiFi-pose work separates fixed-benchmark pose estimation from robustness-focused protocols: Person-in-WiFi 3D uses a fixed receiver layout with receiver-count analysis, while DT-Pose studies domain-consistent representation learning as a dedicated cross-domain problem [35]. We follow the former convention to focus on compact architecture design and application-scale inference.

III. SYSTEM AND PROBLEM FORMULATION

A. Channel State Information

Commodity WiFi devices communicate over N orthogonal subcarriers; for each subcarrier i the received and transmitted symbols are related by

$$Y_i = H_i X_i + n_i, \quad i \in [1, N], \quad (1)$$

where $H_i \in \mathbb{C}$ is the channel state information (CSI), X_i the transmitted symbol, and n_i additive noise. Each H_i encodes the joint effect of path loss, scattering, reflection, and small-scale fading along the propagation path, so its amplitude and phase vary with the body geometry, motion, and pose of people between transmitter and receiver. This dependency makes WiFi a viable sensing modality for pose, but it also means the signal aggregates contributions from the whole scene rather than localizing them to a specific person, which is fundamentally harder than visual input.

B. Phase Sanitization

Raw CSI phase is corrupted by carrier-frequency offset, sampling-frequency offset, and packet detection delay, which together appear as a per-packet linear drift across subcarriers. Standard sanitization techniques fit and remove this drift before the signal is consumed by a learning model [36]. We follow the project default and apply discrete-wavelet amplitude denoising together with linear-phase normalization at the dataset stage; this leaves both amplitude and a stabilized phase available to the downstream pipeline. The denoising stage is treated as a fixed preprocessing component and is not retrained.

C. Hardware Configuration

The sensing layout uses one transmitter and three receivers around a 3.5×4 m indoor area. Each receiver contributes three antenna streams, and each stream provides 30 active subcarriers over a 20-packet temporal window. After amplitude and phase are concatenated, the transmitter dimension is omitted because only one transmitter is used, leaving a $3 \times 3 \times 20 \times 60$ tensor that is flattened into the 180 WiFi tokens consumed by the encoder in Section IV. Section V-A gives the recording protocol.

D. Task Definition

Given a CSI tensor from the fixed transmitter–receiver layout, the model produces 100 score-ranked pose queries, each represented by 14 3D keypoints. Following Person-in-WiFi 3D, evaluation uses Hungarian assignment between the query set and Kinect-derived ground-truth poses. The target is therefore low pose error together with reliable coverage of people under the fixed-layout protocol.

IV. METHOD

A. Overview

Figure 1 summarizes the pipeline. A preprocessed CSI tensor is converted into WiFi tokens, encoded by a compact sequence model, decoded into person hypotheses, and refined from draft pose to final pose through a flow-based correction module.

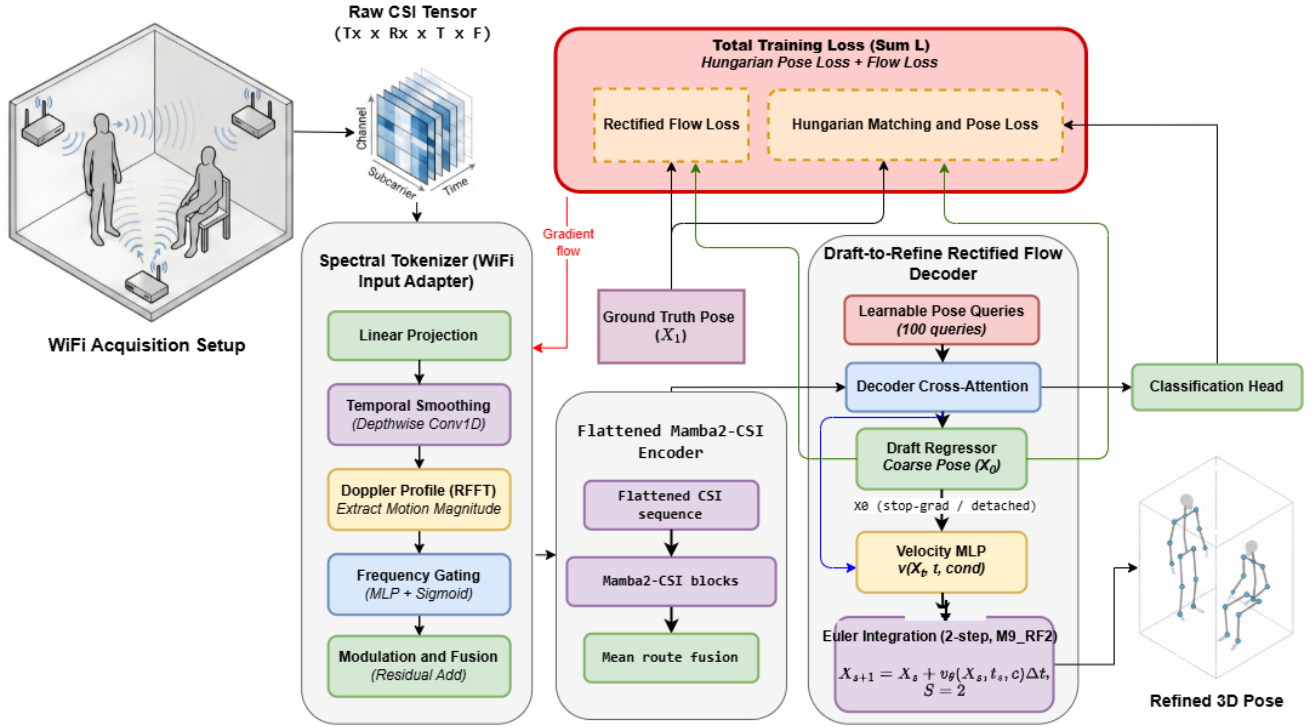


Fig. 1. System overview of the compact draft-to-refine pipeline.

B. Motion-Aware Spectral Tokenization

The baseline maps each 60-dimensional CSI feature to a 256-dimensional token. The spectral tokenizer preserves this linear path as a residual bypass and reshapes the resulting 180 tokens into nine spatial groups with 20 temporal steps each. For every group, a depthwise temporal Conv1D (kernel size 5) followed by GELU smooths local variation. A real FFT along the 20-step axis produces an 11-bin magnitude profile, which a two-layer MLP maps to 20 sigmoid gating weights. The gated temporal features are channel-projected, added to the linear tokens, and normalized with LayerNorm. This branch is motion-aware feature modulation after projection, rather than a direct Doppler estimator.

C. Flattened Mamba-2 Sequence Encoder

The flattened Mamba-2 sequence encoder replaces a Transformer-heavy encoder with a compact sequence model. In the selected configuration, tokens are processed as a single WiFi sequence rather than through multiple manually routed branches. This reduces parameter count and latency while retaining sequence context across spatial and temporal CSI tokens.

D. Lightweight Query-Pose Head and Flow Refinement

The lightweight query-pose head predicts person hypotheses through 100 learned pose queries. During training, Hungarian assignment and the classification, keypoint, and bone losses are computed on draft predictions. For matched positives, the flow module receives a detached 42-dimensional draft pose, the target pose, and a 256-dimensional query condition. The training objective follows a conditional velocity-matching form:

$$x_t = (1 - t)x_0 + tx_1, \quad (2)$$

$$v^*(x_t, t, c) = x_1 - x_0, \quad (3)$$

$$\mathcal{L}_{flow} = \|v_\theta(x_t, t, c) - v^*(x_t, t, c)\|_2^2. \quad (4)$$

The draft is perturbed with Gaussian noise of standard deviation 0.1 during flow training, and t is sampled uniformly from $[0, 1]$. The velocity network concatenates the 42-dimensional pose, the 256-dimensional query condition, and a 64-dimensional sinusoidal time embedding; it projects this 362-dimensional input to 512 dimensions, applies three residual MLP blocks, and predicts a 42-dimensional velocity. At inference, Euler integration updates the pose:

$$x_{k+1} = x_k + \Delta t v_\theta(x_k, t_k, c). \quad (5)$$

The one-step and two-step settings use Euler steps of size $\Delta t = 1$ and $\Delta t = 0.5$, respectively, to report the final 3D keypoints.

Rectified-Flow Architecture

$$\hat{X}_{\text{ref}} = \text{Flow}(X_{\text{draft}}, q_{\text{feat}}, t), \quad v_{\theta} = f_{\theta}(X_t, q_{\text{feat}}, t)$$

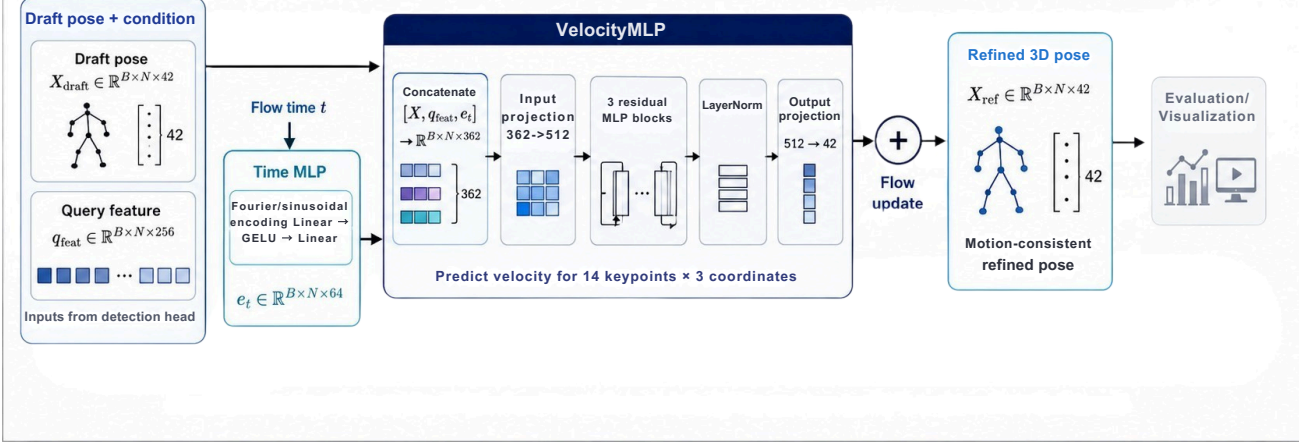


Fig. 2. Conditional pose-flow refinement from draft coordinates to refined 3D keypoints.

E. Implementation Details

The selected Mamba-2 Flow (two steps) configuration uses 100 learned person queries, 256-dimensional embeddings, and 14 output keypoints. CSI is first mapped by a spectral `WifiInputAdapter`; the flattened Mamba-2 sequence encoder has three layers with $d_{\text{state}}=64$, $d_{\text{conv}}=4$, expansion 2, head dimension 64, dropout 0.1, a `time_major` route, mean route fusion, no positional embedding, and no final attention block. The lightweight query-pose head applies 8-head query attention, a two-layer MLP draft regressor, and a binary classification head. The flow refiner is a conditional `VelocityMLP` over 3D keypoint coordinates and query-level features.

Training uses focal classification loss with weight 2.0, keypoint L1 loss with weight 5.0, bone-length loss with weight 2.0, and flow loss with weight 10.0 for the flow-refined variants. All configurations use AdamW with learning rate 2×10^{-5} and weight decay 10^{-4} . At evaluation time, predictions are assigned to ground truth with Hungarian matching under a 500 mm threshold. Unmatched ground-truth persons receive a 500 mm penalty in overall MPJPE; Matched MPJPE excludes this penalty and averages only over assigned prediction-ground-truth pairs.

V. EXPERIMENTAL METHODOLOGY

A. Dataset and Acquisition

We use the publicly released Person-in-WiFi 3D dataset and acquisition protocol [1]. The original dataset paper reports ethics clearance for public release; this study collects no additional participant data. Four ThinkPad X201 laptops equipped with Intel 5300 WiFi adapters form one transmitter and three receivers. The transmitter operates on channel 128 at 5.64 GHz through a single antenna, while each receiver records three antenna streams. It sends 300 packets per second and retains 30 subcarriers per transmitter-receiver link. An Azure Kinect captures synchronized RGB-D observations at 15 frames per second. Each CSI block has size $1 \times 3 \times 3 \times 30 \times 20$ for transmitter, receiver, antenna, subcarrier, and time; concatenating amplitude with sanitized phase, removing the singleton transmitter axis, and reordering axes yields the $3 \times 3 \times 20 \times 60$ input and 180 tokens.

The dataset contains recordings from seven volunteers performing eight daily actions across an office, classroom, and corridor. Each 40-second clip contributes its first 550 frames to training and final 50 frames to testing, following the baseline protocol. Kinect SDK failures, including four-person cases, are excluded in the released benchmark; the resulting 1P/2P/3P split contains 89,946 training and 7,824 test samples, summarized in Table I.

TABLE I
DATASET STATISTICS FOR THE PERSON-IN-WIFI 3D TEST BED USED THROUGHOUT THIS PAPER. COUNTS ARE CSI SAMPLES; EACH SAMPLE IS PAIRED WITH A SYNCHRONIZED AZURE KINECT SKELETON.

split	1-person	2-person	3-person	total
training	28,121	36,242	25,583	89,946
test	2,586	3,184	2,054	7,824
total	30,707	39,426	27,637	97,770

B. Training Setup

All configurations use the same 20-epoch AdamW schedule with base learning rate 2×10^{-5} , weight decay 10^{-4} , batch size 32, deterministic seed 42, and FP32 precision. CSI inputs use discrete-wavelet amplitude denoising and phase normalization. Experiments run on an NVIDIA RTX 6000 Ada Generation GPU with PyTorch 2.1.0, CUDA 12.1, and cuDNN 8.9.2. Runtime is model-only `simple_test` inference on a batch-one $3 \times 3 \times 20 \times 60$ tensor: preprocessing, data I/O, and evaluation matching are excluded. Each result uses 10 warm-up iterations followed by 100 synchronized FP32 measurements; latency is the mean, throughput its reciprocal, and memory is peak CUDA allocated memory.

C. Research Questions

The results are organized around three questions. RQ1 asks how representation, encoder, and compact pose-head choices affect the accuracy–efficiency trade-off. RQ2 asks how flow inference changes matched localization and missed-person behavior. RQ3 asks which configuration provides the most useful fixed-layout operating point under the shared protocol.

D. Evaluation Metrics

For a predicted 3D pose $\hat{P}$ and the matched ground truth P_{gt} , the per-joint position error (PJPE) of joint k is the Euclidean distance between predicted and ground-truth coordinates, and the mean per-joint position error (MPJPE) averages PJPE over the $K=14$ keypoints [37]:

$$\text{PJPE}(k) = \left\| \hat{P}_k - P_{gt,k} \right\|_2, \quad \text{MPJPE} = \frac{1}{K} \sum_{k=1}^K \text{PJPE}(k). \quad (6)$$

To diagnose where error concentrates, we additionally report the per-joint dimension localization error (PJDLE), defined as the ℓ_1 distance along each spatial axis $a \in \{h, v, d\}$:

$$\text{PJDLE}_a(k) = \left| \hat{P}_{k,a} - P_{gt,k,a} \right|, \quad \text{MPJDLE}_a = \frac{1}{K} \sum_{k=1}^K \text{PJDLE}_a(k). \quad (7)$$

For every frame, the 100 score-ranked queries are assigned to ground truth by Hungarian matching. Pairs above 500 mm are rejected; each unmatched ground-truth person contributes a 500 mm penalty to overall MPJPE. Thus, overall MPJPE combines matched-pose localization and missed-person coverage, whereas Matched MPJPE averages only accepted pairs. This inherited metric is used consistently for all variants; it is not a detection average-precision measure. Results are further separated by scene cardinality (1P/2P/3P).

VI. RESULTS

A. Main Ablation

RQ1 is answered by the full architecture ladder. Table II maps stable experiment IDs to functional public names, while Table III reports the corresponding accuracy and efficiency. The PETR variants vary the input and encoder; the draft variants introduce the lightweight query-pose head without refinement; and the flow variants add one- or two-step correction under Transformer, Mamba, and flattened Mamba-2 sequence encoders.

The main observation is that efficiency improves before accuracy reaches its best point. Mamba-2 Draft is the throughput and memory endpoint, reaching 247.7 FPS with 1.813M parameters and 18.41 MB peak allocated memory. Mamba-2 Flow (two steps) improves MPJPE by 7.053 mm (4.09%) over PETR Reference and by 2.599 mm over Mamba-2 Flow (one step). It reaches 209.9 FPS with 3.618M parameters and 25.29 MB peak allocated memory, compared with 118.7 FPS, 13.130M parameters, and 155.60 MB for PETR Reference. This corresponds to $1.77 \times$ higher throughput and 72.45% fewer parameters while delivering the lowest MPJPE in the evaluated ladder.

Figure 3 visualizes the same log as a compact multi-panel accuracy–efficiency summary. The appendix retains the complementary bubble view.

TABLE II
CONTROLLED ARCHITECTURE LADDER. EXPERIMENT IDS PROVIDE STABLE LINKS TO BENCHMARK ARTIFACTS; PUBLIC LABELS DESCRIBE THE EVALUATED ARCHITECTURE.

Experiment ID	Input	Encoder	Pose Head	Refinement	Public label
■ M0	Linear	Transformer	PETR query decoder	None	PETR Reference
■ M1	Spectral	Transformer	PETR query decoder	None	Spectral PETR
■ M2	Spectral	Mamba	PETR query decoder	None	Mamba PETR
■ M3	Spectral	Flattened Mamba-2	PETR query decoder	None	Mamba-2 PETR
■ M4	Spectral	Transformer	Lightweight query-pose	None	Transformer Draft
■ M5	Spectral	Mamba	Lightweight query-pose	None	Mamba Draft
■ M6	Spectral	Flattened Mamba-2	Lightweight query-pose	None	Mamba-2 Draft
■ M7	Spectral	Transformer	Lightweight query-pose	One step	Transformer Flow
■ M8	Spectral	Mamba	Lightweight query-pose	One step	Mamba Flow
■ M9	Spectral	Flattened Mamba-2	Lightweight query-pose	One step	Mamba-2 Flow (1 step)
■ M9_RF2	Spectral	Flattened Mamba-2	Lightweight query-pose	Two steps	Mamba-2 Flow (2 steps)

TABLE III
FULL 20-EPOCH ABLATION. LOWER MPJPE, PARAMETER COUNT, AND MEMORY ARE BETTER; HIGHER FPS IS BETTER.

Experiment ID	MPJPE	1P	2P	3P	FPS	Params (M)	Memory (MB)
■ M0	172.540	136.141	167.971	192.537	118.7	13.130	155.60
■ M1	169.271	132.597	163.323	190.810	83.1	13.199	155.86
■ M2	174.961	143.475	166.642	196.772	69.4	11.967	151.16
■ M3	175.658	140.116	169.358	197.084	102.0	10.292	142.69
■ M4	169.796	126.479	167.593	190.251	180.8	5.255	31.61
■ M5	171.055	127.258	171.979	188.481	116.8	4.023	26.80
■ M6	167.573	125.532	164.528	188.363	247.7	1.813	18.41
■ M7	168.342	127.129	162.775	191.390	145.8	7.059	38.49
■ M8	169.653	122.253	165.397	193.943	109.4	5.827	33.68
■ M9	168.086	125.958	161.832	192.229	206.9	3.618	25.29
■ M9_RF2	165.487	121.785	159.341	190.179	209.9	3.618	25.29

TABLE IV
FLOW-SOLVER ANALYSIS. LOWER MPJPE AND MATCHED-ONLY MPJPE ARE BETTER.

Experiment ID	Public setting	MPJPE	Matched	Missed	1P	2P	3P
■ M6	Mamba-2 Draft	167.573	166.381	54	125.532	164.528	188.363
■ M9_no_flow	Mamba-2 Flow (refinement bypassed)	167.732	165.919	82	122.870	163.026	191.421
■ M9	Mamba-2 Flow (1 step)	168.086	166.409	76	125.958	161.832	192.229
■ M9_RF2	Mamba-2 Flow (2 steps)	165.487	163.886	72	121.785	159.341	190.179

B. Flow Solver Analysis

RQ2 separates draft inference, refinement bypass, one-step refinement, and two-step refinement. Table IV and Figure 4 expose two effects: matched-pose localization and unmatched-person count. Mamba-2 Draft produces the fewest unmatched persons, while Mamba-2 Flow (two steps) attains the lowest overall and matched-only MPJPE. The two-step solver is therefore the accuracy-oriented inference setting for the Mamba-2 Flow configuration.

C. Operating Point and Qualitative Cases

RQ3 asks which tested configuration provides the most useful fixed-layout operating point. Mamba-2 Draft is the efficiency endpoint, whereas Mamba-2 Flow (two steps) is the accuracy-oriented choice because it attains the lowest MPJPE while retaining most of the draft model’s speed and size advantages. Figure 5 shows representative 1-person, 2-person, and 3-person

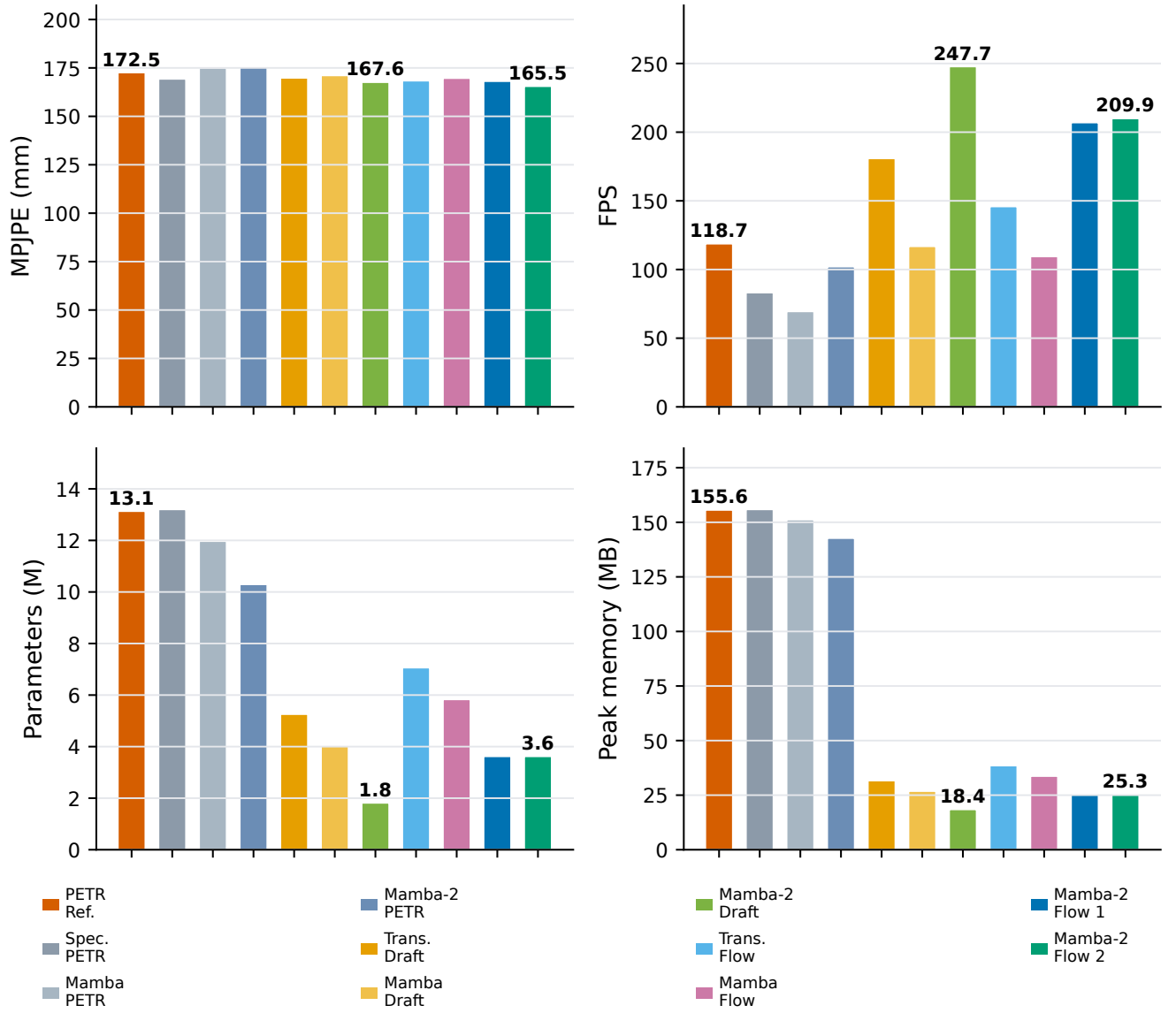


Fig. 3. Accuracy and efficiency across the controlled architecture ladder. Functional labels are defined in Table II.

samples for PETR Reference, Transformer Flow, and Mamba-2 Flow (two steps); quantitative conclusions remain grounded in Tables III and IV.

VII. DISCUSSION

The ablation shows how a compact draft-to-refine design changes the WiFi pose-estimation trade-off. Mamba-2 Draft establishes the speed and memory advantages of the flattened Mamba-2 sequence path, while Mamba-2 Flow (two steps) adds lightweight correction to obtain the lowest MPJPE in the evaluated architecture ladder.

The design ladder also shows that encoder substitution alone is insufficient: the Mamba PETR and Mamba-2 PETR variants do not improve accuracy, whereas the flattened Mamba-2 sequence encoder becomes effective with the lightweight query-pose head and flow refinement. Mamba-2 Draft and Mamba-2 Flow (two steps) consequently serve complementary deployment goals: the former maximizes speed and memory efficiency, while the latter minimizes MPJPE within the measured trade-off.

The complementary diagnostics sharpen this result. Mamba-2 Draft attains fewer unmatched persons and higher throughput, whereas Mamba-2 Flow (two steps) improves overall and matched-only MPJPE. The appendix reports per-joint behavior and a baseline-compatible receiver-masking diagnostic for the intended three-receiver input.

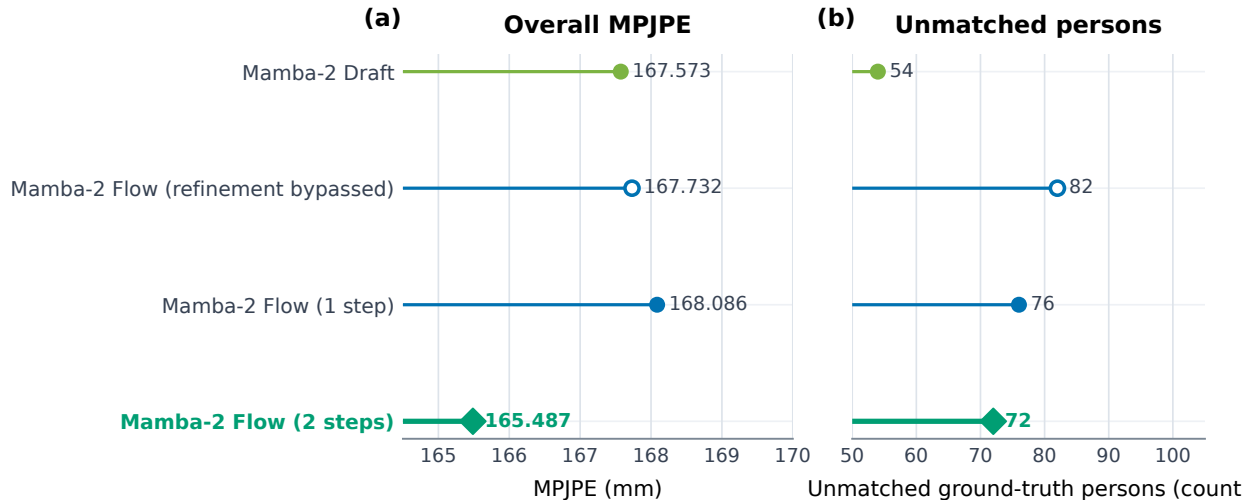


Fig. 4. Flow-solver analysis across four inference settings. Lower values are better in both panels: (a) overall MPJPE and (b) unmatched ground-truth person count. The highlighted point denotes Mamba-2 Flow (2 steps).

TABLE V
ABLATION SUMMARY ACROSS THE EVALUATED ARCHITECTURE LADDER.

Design question	Finding	Evidence from the architecture ladder
Spectral representation	Improves PETR accuracy	Spectral PETR improves over PETR Reference in MPJPE, with a runtime cost.
Efficient sequence modeling	Depends on pose-head design	Mamba-family encoders reduce cost in the lightweight query-pose path, while the PETR variants do not improve accuracy.
Flow correction	Improves the flow setting	Mamba-2 Flow (2 steps) improves over one-step refinement and refinement bypass in overall and matched-only MPJPE.
Operating point	Accuracy-oriented choice	Mamba-2 Flow (2 steps) gives the lowest MPJPE while retaining large speed, parameter, and memory gains over PETR Reference.

VIII. EVALUATION SCOPE

The controlled ablations use deterministic seed 42 under the fixed three-receiver Person-in-WiFi 3D protocol. The reported comparisons therefore characterize architecture trade-offs within this shared sensing layout and temporal split; they do not make cross-environment or random-initialization variance claims.

IX. CONCLUSION

This paper presents a compact draft-to-refine WiFi 3D pose pipeline. Under the inherited Person-in-WiFi 3D protocol, Mamba-2 Flow (two steps) achieves 165.487 mm MPJPE, improves on PETR Reference by 7.053 mm, reaches 209.9 FPS, reduces parameters to 3.618M, and lowers peak allocated memory to 25.29 MB. The architecture ladder shows that a flattened Mamba-2 sequence encoder with lightweight two-step flow refinement delivers the lowest measured MPJPE while retaining substantial efficiency gains over the PETR reference.

DATA AND CODE AVAILABILITY

This work reuses public Person-in-WiFi 3D data [1]. The supplementary material provides the LaTeX source, configuration files, benchmark summaries, and figure scripts for the reported tables and figures.

ETHICS STATEMENT

The Person-in-WiFi 3D dataset used in this study was reported by its creators as ethically cleared for public release [1]. We reuse that released dataset and collect no new human-subject data. RGB-D annotations are used only as synchronized pose supervision within the benchmark protocol.

AUTHOR CONTRIBUTIONS

Nguyen Vinh Nghi, Le Ngoc Anh Thu, and Nguyen Ngo Nhat Nam contributed to system design, experiment execution, analysis, and manuscript preparation. Tran Ngoc Hoang supervised the research direction, reviewed the experimental interpretation, and advised the manuscript preparation.

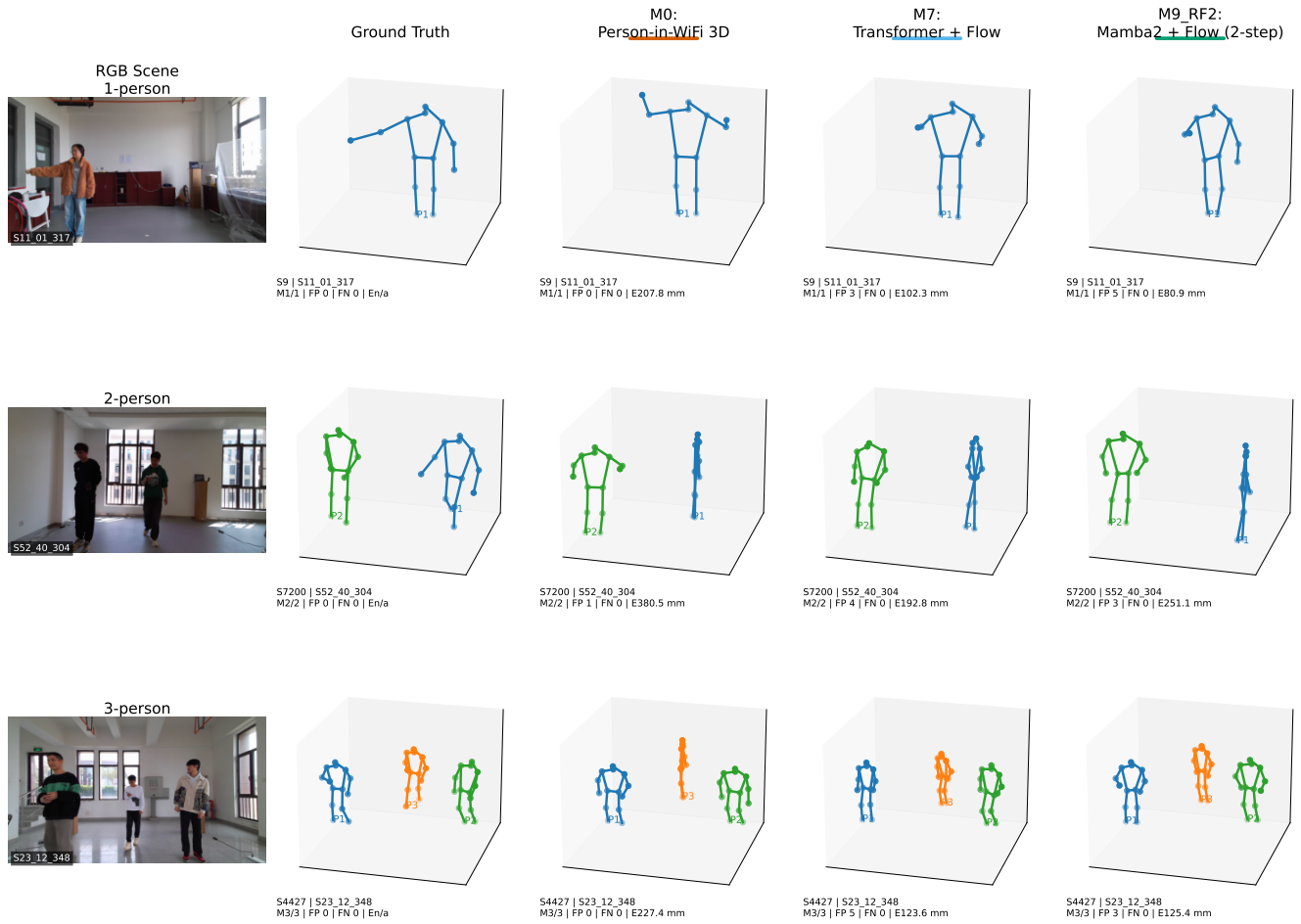


Fig. 5. Selected qualitative WiFi pose samples across 1-person, 2-person, and 3-person settings. Columns show the synchronized RGB scene, Kinect ground truth, PETR Reference, Transformer Flow, and Mamba-2 Flow (two steps) predictions. Unmatched predictions above the fixed confidence threshold are gray. Panel footers report per-sample matching diagnostics rather than aggregate test metrics.

AI DISCLOSURE

This manuscript was prepared with the assistance of AI-powered academic writing tools, in compliance with the ResFes 2026 commitment on responsible use of AI. AI assistance covered structure planning, result summarization, figure and table integration, and LaTeX formatting. All claims, numerical results, and conclusions were directed and reviewed by the authors, who take full responsibility for the accuracy and integrity of the work. No part of the experimental data, trained models, or reported metrics was generated by an AI system.

ACKNOWLEDGMENT

This work builds on the open Person-in-WiFi 3D project and RESFES 2026 ablation experiments.

APPENDIX A SUPPLEMENTARY FIGURES

The following view reuses the full ablation log from Table III to show the accuracy–efficiency trade-off.

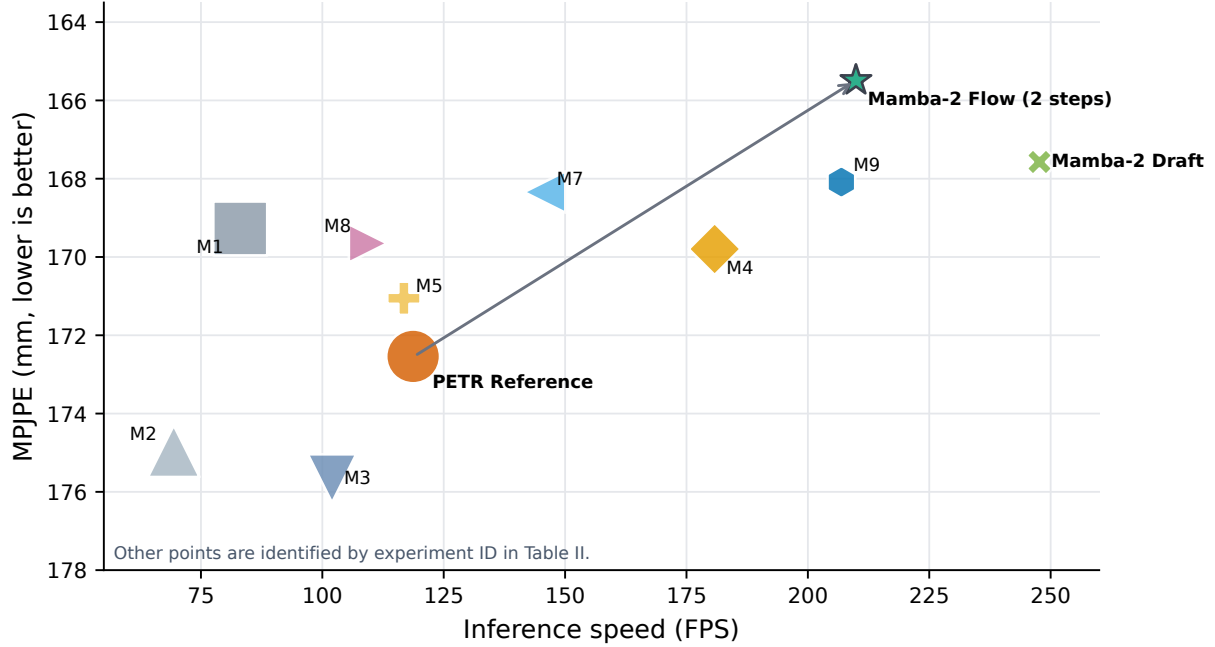


Fig. 6. Accuracy-efficiency trade-off of the evaluated variants. The three functional anchor labels identify the reference, throughput endpoint, and accuracy-oriented setting; other points use stable experiment IDs defined in Table II. Bubble area corresponds to the number of parameters in millions.

APPENDIX B PER-JOINT AND RECEIVER DIAGNOSTICS

Table VI compares Mamba-2 Flow (two steps) with PETR Reference at the joint level.

TABLE VI
PER-JOINT MPJPE FOR MAMBA-2 FLOW (TWO STEPS) VERSUS PETR REFERENCE, EVALUATED ON THE SAME HELD-OUT EVALUATION SET AS TABLE III. LOWER IS BETTER. UNITS: MILLIMETERS.

joint	Mamba-2 Flow (2 steps)	PETR Reference
neck	159.2	163.0
head	147.3	154.2
left shoulder	155.8	162.3
right shoulder	155.2	161.8
left elbow	126.9	139.3
right elbow	193.5	194.0
left wrist	128.7	138.1
right wrist	260.1	251.7
left hip	128.0	146.7
right hip	199.9	203.2
left knee	126.4	139.6
right knee	124.7	142.2
left ankle	132.9	141.6
right ankle	278.3	278.0
Mean	165.5	172.6

APPENDIX C RECEIVER-SUBSET DIAGNOSTIC

Table VII reports the selected model under receiver masking. The full three-receiver input remains the intended fixed-layout operating condition; subset rows are included as a compact sensitivity diagnostic.

TABLE VII

PER-RECEIVER EVALUATION OF MAMBA-2 FLOW (TWO STEPS) ON THE SAME HELD-OUT EVALUATION SET AS TABLE III. RECEIVERS NOT IN THE SUBSET ARE ZERO-MASKED AT THE INPUT OF THE POSE-ESTIMATION PIPELINE (THE MODEL IS NOT RETRAINED). MIRRORS BASELINE TABLE 5 OF [1]. UNITS: MM; LOWER IS BETTER.

receiver subset	MPJPE	1P	2P	3P	matched	missed
3 receivers (R1+R2+R3)	165.5	121.8	159.3	190.2	15044	72
R1+R2	401.6	382.6	407.1	403.9	9567	5549
R1+R3	414.1	431.2	414.7	406.2	8838	6278
R2+R3	433.3	417.2	433.4	440.0	7828	7288
R2 only	498.0	496.3	499.6	497.0	498	14618

APPENDIX D FLOW-LOSS SENSITIVITY

Table VIII reports the two low-flow-loss control runs. These controls are supplementary to the same-configuration solver analysis in Table IV.

TABLE VIII
LOW-FLOW-LOSS CONTROL RUNS. LOWER MPJPE AND MATCHED-ONLY MPJPE ARE BETTER.

Inference	MPJPE	Matched	Missed	1P	2P	3P
Two steps	167.981	166.082	86	121.335	160.970	194.804
Four steps	169.454	167.275	99	120.505	161.851	197.854

REFERENCES

- [1] K. Yan, F. Wang, B. Qian, H. Ding, J. Han, and X. Wei, "Person-in-wifi 3d: End-to-end multi-person 3d pose estimation with wi-fi," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [2] C. Zheng, W. Wu, T. Yang, S. Zhu, C. Chen, R. Liu, J. Shen, N. Kehtarnavaz, and M. Shah, "Deep learning-based human pose estimation: A survey," *ACM Computing Surveys*, vol. 56, no. 1, pp. 1–37, 2023.
- [3] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, "Realtime multi-person 2d pose estimation using part affinity fields," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017.
- [4] H.-S. Fang, S. Xie, Y.-W. Tai, and C. Lu, "RMPE: Regional multi-person pose estimation," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2334–2343.
- [5] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 5693–5703.
- [6] M. Zhao, T. Li, M. Abu Alsheikh, Y. Tian, H. Zhao, A. Torralba, and D. Katabi, "Through-wall human pose estimation using radio signals," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7356–7365.
- [7] F. Adib, H. Mao, Z. Kabelac, D. Katabi, and R. C. Miller, "Smart homes that monitor breathing and heart rate," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2015, pp. 837–846.
- [8] H. Abdelnasser, M. Youssef, and K. A. Harras, "WiGest: A ubiquitous Wi-Fi-based gesture recognition system," in *Proceedings of the IEEE Conference on Computer Communications*, 2015, pp. 1472–1480.
- [9] Y. Zhang, Y. Zheng, K. Qian, G. Zhang, Y. Liu, C. Wu, and Z. Yang, "Widar3.0: Zero-effort cross-domain gesture recognition with Wi-Fi," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 8671–8688, 2022.
- [10] F. Wang, J. Han, S. Zhang, X. He, and D. Huang, "CSI-Net: Unified human body characterization and pose recognition," *arXiv preprint arXiv:1810.03064*, 2018.
- [11] F. Wang, S. Panev, Z. Dai, J. Han, and D. Huang, "Can WiFi estimate person pose?" *arXiv preprint arXiv:1904.00277*, 2019.
- [12] F. Wang, S. Zhou, S. Panev, J. Han, and D. Huang, "Person-in-WiFi: Fine-grained person perception using WiFi," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019.
- [13] W. Jiang, H. Xue, C. Miao, S. Wang, S. Lin, C. Tian, S. Murali, H. Hu, Z. Sun, and L. Su, "Towards 3D human pose construction using WiFi," in *Proceedings of the 26th Annual International Conference on Mobile Computing and Networking (MobiCom)*, 2020.
- [14] Y. Ren, Z. Wang, S. Tan, Y. Chen, and J. Yang, "WiNect: 3D human pose tracking for free-form activity using commodity WiFi," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 5, no. 4, pp. 1–29, 2021.
- [15] Y. Ren, Z. Wang, Y. Wang, S. Tan, Y. Chen, and J. Yang, "GoPose: 3D human pose estimation using WiFi," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 6, no. 2, pp. 1–25, 2022.
- [16] J. Yang *et al.*, "MM-Fi: Multi-modal non-intrusive 4D human dataset for versatile wireless sensing," in *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.
- [17] J. Yang, Y. Zhou, H. Huang, H. Zou, and L. Xie, "MetaFi: Device-free pose estimation via commodity WiFi for metaverse avatar simulation," in *IEEE World Forum on Internet of Things*, 2022.
- [18] H. Boudlal, M. Serrhini, and A. Tahiri, "A novel approach for simultaneous human activity recognition and pose estimation via skeleton-based leveraging WiFi CSI with YOLOv8 and MediaPipe frameworks," *Signal, Image and Video Processing*, vol. 18, pp. 3673–3689, 2024.
- [19] C. Yin, X. Miao, J. Chen, H. Jiang, J. Yang, Y. Zhou, M. Wu, and Z. Chen, "PowerSkel: A device-free framework using CSI signal for human skeleton estimation in power station," *IEEE Internet of Things Journal*, vol. 11, pp. 20 165–20 177, 2024.
- [20] J. Chen, Y. Qu, R. Tang, and D. Slock, "Graph-based 3D human pose estimation using WiFi signals," *arXiv preprint arXiv:2511.19105*, 2025.
- [21] Y. Dao, L. Zhang, H. Liu, H. Zhang, and W. Wang, "WiFlow: A lightweight Wi-Fi-based continuous human pose estimation network with spatio-temporal feature decoupling," *arXiv preprint arXiv:2602.08661*, 2026.

- [22] Y. Ren, Z. Wang, Y. Wang, S. Tan, Y. Chen, and J. Yang, "3D human pose estimation using WiFi signals," in *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*, 2021.
- [23] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017.
- [24] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *European Conference on Computer Vision (ECCV)*, 2020, pp. 213–229.
- [25] D. Shi, X. Wei, L. Li, Y. Ren, and W. Tan, "End-to-end multi-person pose estimation with transformers," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 11 069–11 078.
- [26] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Conference on Language Modeling*, 2024.
- [27] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," *arXiv preprint arXiv:2401.09417*, 2024.
- [28] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, and Y. Liu, "VMamba: Visual state space model," *arXiv preprint arXiv:2401.10166*, 2024.
- [29] T. Huang, X. Pei, S. You, F. Wang, C. Qian, and C. Xu, "LocalMamba: Visual state space model with windowed selective scan," *arXiv preprint arXiv:2403.09338*, 2024.
- [30] T. Dao and A. Gu, "Transformers are ssms: Generalized models and efficient algorithms through structured state space duality," in *International Conference on Machine Learning*, 2024.
- [31] X. Liu, C. Gong, and Q. Liu, "Flow straight and fast: Learning to generate and transfer data with rectified flow," in *International Conference on Learning Representations*, 2023.
- [32] J. Gong, L. G. Foo, Z. Fan, Q. Ke, H. Rahmani, and J. Liu, "DiffPose: Toward more reliable 3D pose estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13 041–13 051.
- [33] W. Shan, Z. Liu, X. Zhang, Z. Wang, K. Han, S. Wang, S. Ma, and W. Gao, "Diffusion-based 3D human pose estimation with multi-hypothesis aggregation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 14 715–14 725.
- [34] H. Kang, Y. Wang, M. Liu, D. Wu, P. Liu, X. Yuan, and W. Yang, "Diffusion-based pose refinement and multi-hypothesis generation for 3D human pose estimation," in *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2024, pp. 5130–5134.
- [35] Y. Chen, J. Guo, S. Guo, J. Zhou, and D. Tao, "Towards robust and realistic human pose estimation via WiFi signals," *arXiv preprint arXiv:2501.09411*, 2025.
- [36] S. Sen, B. Radunovic, R. R. Choudhury, and T. Minka, "You are facing the Mona Lisa: Spot localization using PHY layer information," in *Proceedings of the 10th International Conference on Mobile Systems, Applications, and Services (MobiSys)*, 2012, pp. 183–196.
- [37] C. Ionescu, D. Papava, V. Olaru, and C. Sminchisescu, "Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 7, pp. 1325–1339, 2014.